

A No-Code Versatile XR Framework with Shared Experiences

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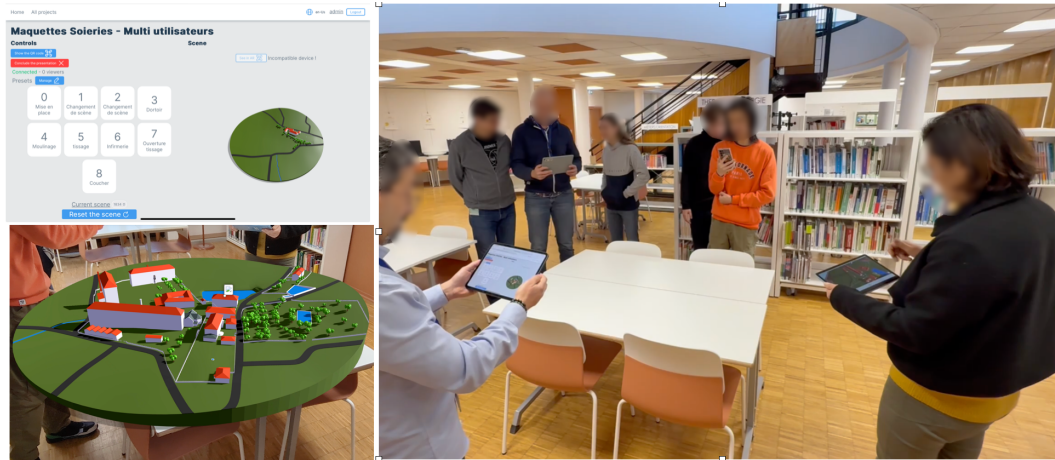


Figure 1: HERA in action. Left: Master and Viewing interfaces. Right: multiple users sharing the Soieries Bonnet experience.

ABSTRACT

This paper introduces the new version of HERA, a no-code versatile open-source XR Authoring framework with shared user experiences. HERA allows to easily create XR applications without any technical knowledge. It is entirely web-based and therefore does not need any installation. It uses well known technologies and is compatible with any WebXR device. The new version presented in this paper allows multi-user XR experience sharing: it proposes a companion application used to remotely control and present a project to several people at the same time. To illustrate this feature, we present two multi-user XR demos: a virtual historical model AR showcase, and a VR environment exploration.

Index Terms: XR Authoring, Cultural heritage, Multi-user, Open-source.

1 INTRODUCTION

XR technology offers significant advantages for cultural heritage: it enhances user engagement and facilitates effective mediation [2], [3]. However, the creation of XR applications can be complex. Therefore, the development of a no-code XR platform is essential to simplify application creation and dissemination. Such a platform would be beneficial for curators and administrators, enabling them to create and maintain dedicated XR experiences without requiring technical expertise. It would also be advantageous for educators, providing immersive learning applications. To address these needs,

we propose a new version of HERA [1], an open-source XR application framework that enables no-code application creation. This new version is versatile, supporting both AR and VR applications and operating on any WebXR-compatible phone, tablet or dedicated XR devices such as the Meta Quest 3 and Apple Vision Pro. It also allows the creation of multi-user applications, with a master user controlling the virtual scene and multiple clients sharing the same experience. This demonstration showcases these new features.

2 HERA FRAMEWORK PRESENTATION

HERA comprises two distinct components: the Authoring Editor and the XR Viewer.

The Authoring Editor application facilitates the creation of scenes. These scenes can be viewed either in augmented reality (AR) or virtual reality (VR) and are composed of imported assets organized in space and time. HERA allows importing and editing various types of assets, including animated 3D meshes, pictures, and text labels. The Editor includes an integrated 3D interface that simplifies the placement and editing of assets. Please note that, at present time, HERA does not allow the creation of 3D assets, external applications like 3D scanners or computer graphics suites must be used for that. Once a project is finalized, it may be published and accessed with the Viewer.

The XR Viewing application enables users to visualize scenes created with the Editor. Users can connect to a dedicated URL to view available projects. Upon selection and loading, a chosen project can be visualized in either AR or VR, depending on its type.

The main addition to this version of HERA is multi-user experience sharing: if an authorized user logs in with the appropriate credentials in the XR viewing application, he becomes the "master", can initiate presentations and therefore controls the same project viewed by multiple users, as shown on the right of figure 1. This process involves sharing a QR code with interested users, which allows them to connect to the presentation. They all see the same scenes, at the same time (bottom left on figure 1). The available

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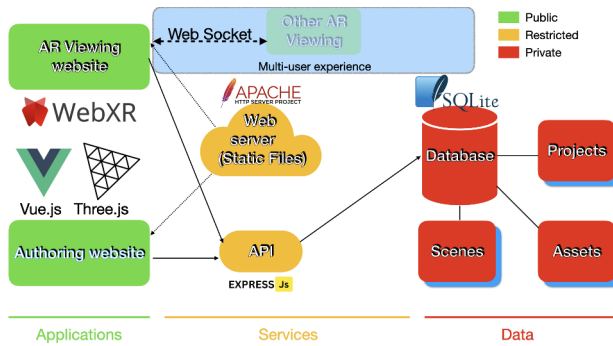


Figure 2: HERA's software architecture

bandwidth is the only limitation on the number of users sharing experiences. Users may even be in different locations. The master has a dedicated web interface in the viewer, similar to a companion application, and may define presets for complex interactions (top left on figure 1). This interface may even be used in a mono-user context: in this case, with a smartphone in hand, the user will be able to operate it as a control interface, allowing him to trigger animations, display assets, and change scene in the application.

Both the Editor and Viewer applications are web-based and do not require any installation. The Editor operates on any Three.js-compatible device, while the Viewer operates on any WebXR compatible device (mainly Android smartphones and tablets, and XR headsets).

3 HERA ARCHITECTURE AND IMPLEMENTATION DETAILS

As seen on figure 2, the architecture is segmented into three levels: Applications, Services and Data, each of these having different access privileges.

1. The **Data level** holds multimedia content: 3D models, annotations, and any other resources that creators may want to integrate into their AR experiences. It is implemented with a *SQLite* database. For security concerns, this level is private and is only accessible via the API (detailed below).
2. The **Service level** is a restricted level containing services needed to access data and deploy the applications. It may only be accessed by the technical administrator of the system. It contains a Web Server and an access API:

A Web Server is needed to deploy the two user applications. The server needs to be able to handle *node.js*. We chose Apache in our own implementation.

The API acts as a bridge that allows the applications to access, add, or remove elements in the database. It has been implemented using *Express.js*.

3. The **Applications level** is public and offers access to the two user applications, the Viewer and the Editor. Both have been implemented using *Vue.js*, *Three.js* and *WebXR*.

As said in the previous section, scene viewing may be shared among multiple users. In this case, the viewing applications of all users communicate using web sockets with the WSS protocol. This protocol is standard.

4 DEMOS

We present two demonstrations created with HERA to illustrate its multi-user capabilities: an augmented reality guided model showcase, and an asymmetric virtual reality environment exploration.

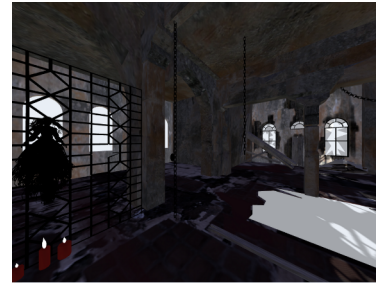


Figure 3: Haunted house demo

The AR demonstration, shown in the video, enables visitors to observe the evolution of the *Soieries Bonnet*, in Jujurieux (F), over two centuries. When set in the year 1870, it also depicts, using animations and labels, a typical day from a worker in the Factory. The demonstration is presented as a virtual model of the site, visualized by multiple visitors connected on the same presentation. The master user dynamically controls the scene display and changes the date, highlights specific buildings, and triggers animations, acting like a guide in a museum visit.

The VR demonstration, shown on figure 3, is an asymmetric application about visiting a Haunted Mansion. It involves two participants, a visitor exploring the mansion and a master who controls the environment. The visitor is immersed in the mansion, while the master sees what the visitor sees and, similarly to a game master in a role playing game, triggers assets and animations according to the visitor's actions.

5 CONCLUSION AND PERSPECTIVES

In this paper we presented demos showcasing the latest iteration of HERA, a versatile no-code authoring framework for XR. Designed for ease of use and adaptability, HERA easily adapts to diverse applications. As an open-source platform, it offers customization options to meet specific requirements. It also allows tailored companion applications through shared experiences. Interactions are facilitated via HTTP API requests, enabling seamless integration into web or native applications.

Future enhancements include advanced interaction design features such as spatialized triggers. Integration of audio elements within scenes is also planned. Additionally, user study tools will be developed to capture user interactions and traces.

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